

APPLIED DATA SCIENCE PROJECT

DEEP: Dupilumab Eosinophilia Early Prediction

Politecnico di Torino

FONDAZIONE links PASSION FOR INNOVATION

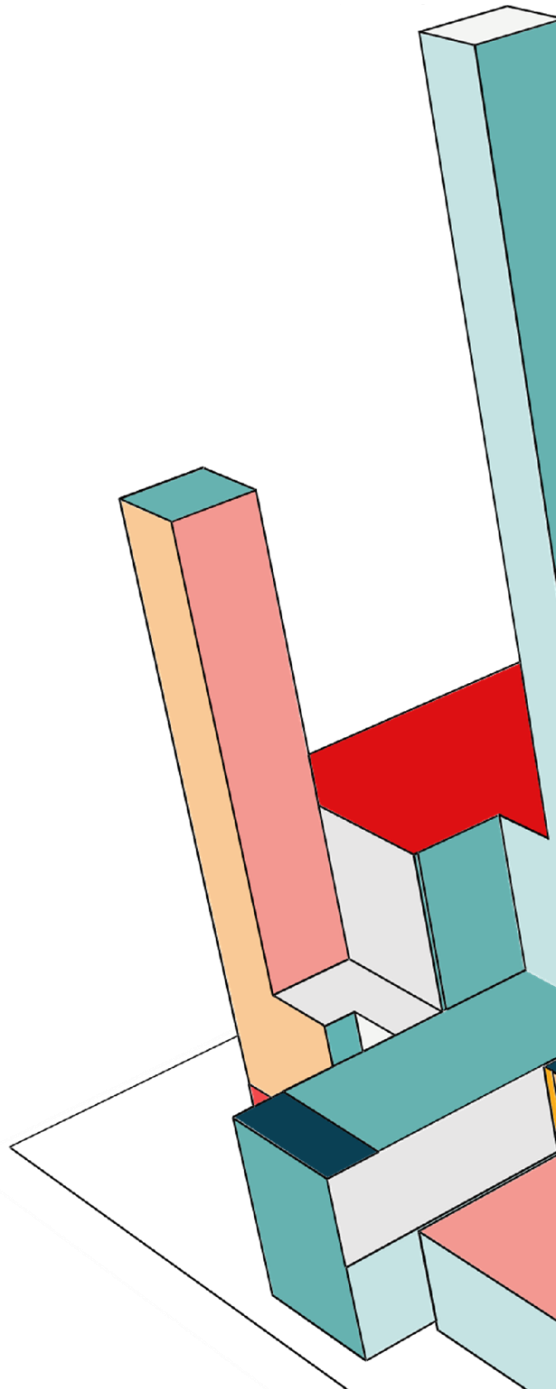
ellis European Laboratory for Learning and Intelligent Systems

VALUE-DRIVEN PROJECT

In precision medicine, timely and accurate prediction of therapy response is crucial to tailor personalized treatment. This is especially critical for biological therapies such as **Dupilumab**, the first biologic treatment approved in Italy (2020), for Chronic Rhinosinusitis with Nasal Polyps (CRSwNP). Although effective, Dupilumab treatment seems to be associated with **an increase in blood eosinophils**, leading clinicians to discontinue treatment or switch to an alternative biologic.

Today it is not possible to predict which patients will develop this condition and at what stage of treatment. Identifying them in advance would let doctors choose the most suitable biologic therapy, improving treatment management and supporting hospital resource planning.

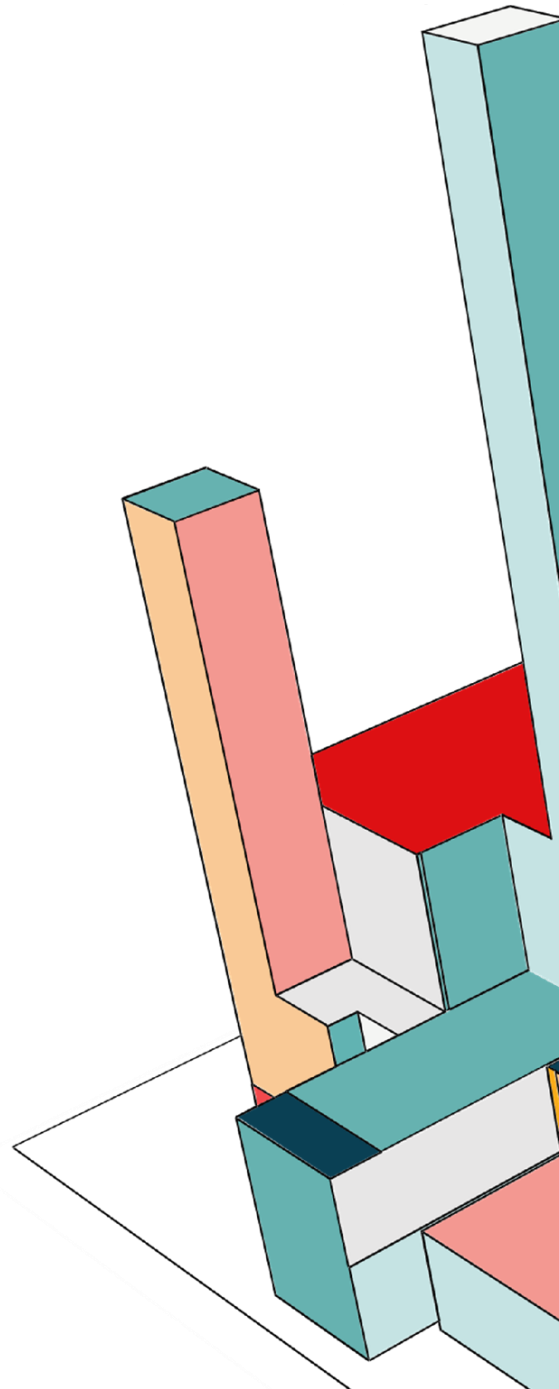
The project is supervised by **LINKS Foundation** in collaboration with the ENT Ward of Molinette Hospital



DATA

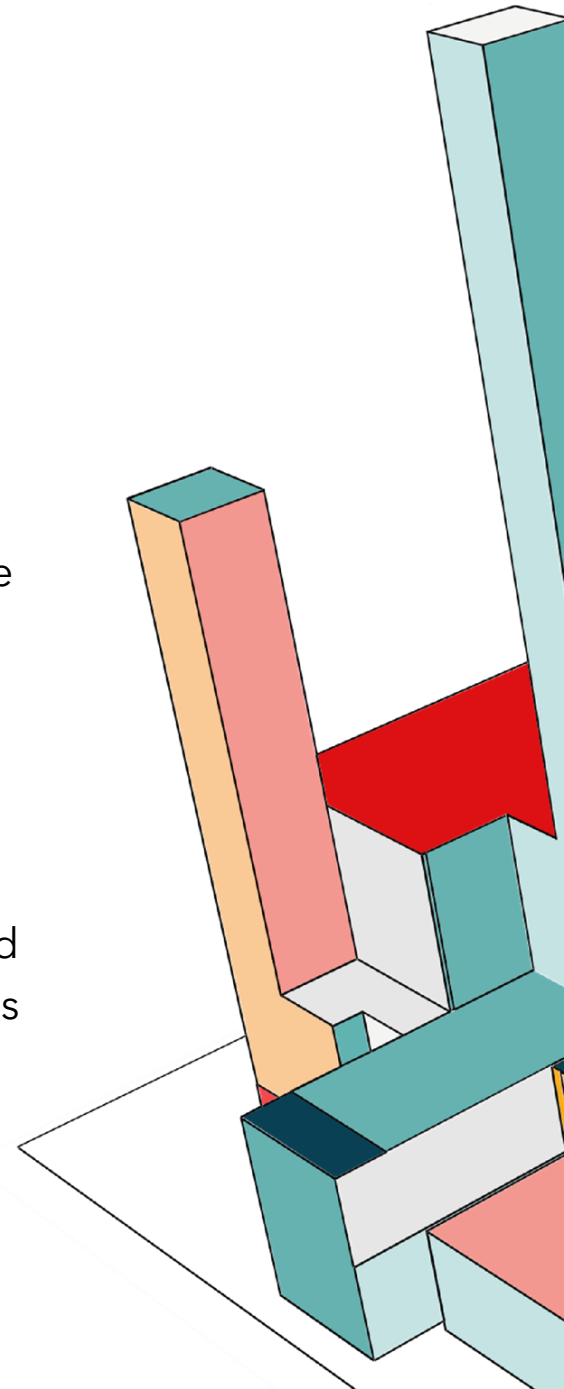
- Data from 110+ patients with CRSwNP that underwent a Dupilumab treatment, from 2021 to today. The dataset contains their demographic and clinical information, together with reports of their first visit and follow-up visits up to 3 years of treatment.

- Data has been manually collected by medical doctors in ENT ward of Molinette hospital. The use of this data for research purposes has been approved by the hospital's Ethics Committee.

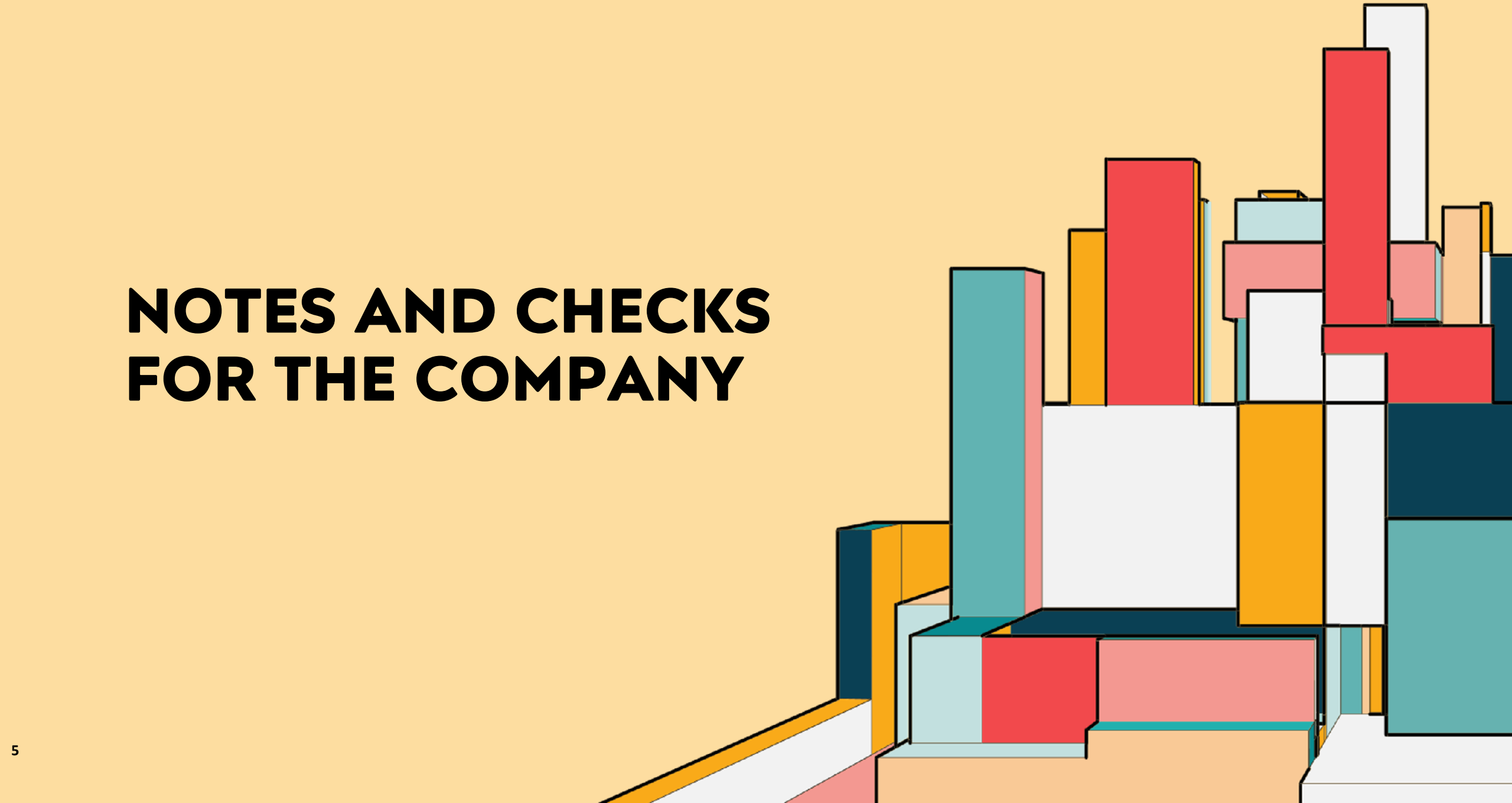


TASK

- The main task of this project is to assess whether baseline demographic and clinical features can identify patients at increased risk of exceeding a specified threshold of blood eosinophil count ($0.8 \times 10^3/\mu L$) .
- This task will require a first data cleaning and data exploration phase, followed by a feature selection process and the training and validation of prediction model(s).
- The specific approaches (supervised/unsupervised) and models applied in this task (e.g. statistical regression or machine learning algorithms) will be decided along the course of the project, possibly with a comparison study between the different approaches.
- Given the nature of the dataset, that contains repeated eosinophil measurements collected at different time points, an optional research question could be predicting when a patient's eosinophil count is likely to exceed the predefined threshold, or even predicting the treatment response.



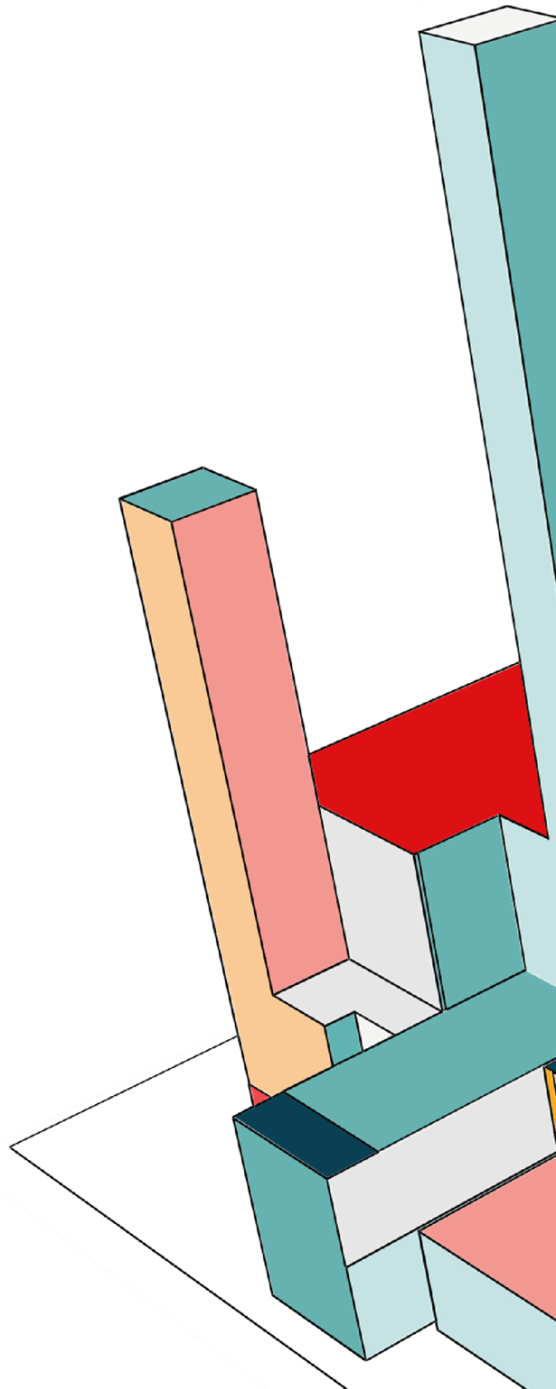
NOTES AND CHECKS FOR THE COMPANY



LIGHT MENTORING

- **Mentors**
  - Angelica Urbanelli (angelica.urbanelli@linksfoundation.com)
  - Sara De Luca (sara.deluca@linksfoundation.com)

- **Weekly one-hour calls** with students for the whole duration of the project
- Feel free to reach out via **Teams** or **email** for any questions or doubts
- Dataset provided by the company



POLICY

- Both project descriptions and implementations will be part of a repository group published on GitHub

- The repositories will be public

- The data provided **must not be shared** and/or **published**
- The provided data **must not** be disclosed to, uploaded to, or processed by **any external LLM-based platform or AI service**, including via API calls, third-party integrations, or other remote interfaces

